

A Single Stepsize Suffices for Unprojected Linear TD(0)

Simultaneous Robust and Fast Rates via Polyak–Ruppert Averaging

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Setting I: from RL successes to policy evaluation



- **State** s_t : what the agent sees now.
- **Action** a_t : what the agent does next.
- **Policy** μ : a rule mapping states to actions.
- **Reward** r_t : the immediate feedback received after taking an action.

AlphaGo's success

- Once a policy is fixed, we want to know quickly: *how good is this policy from each state?*

Policy evaluation

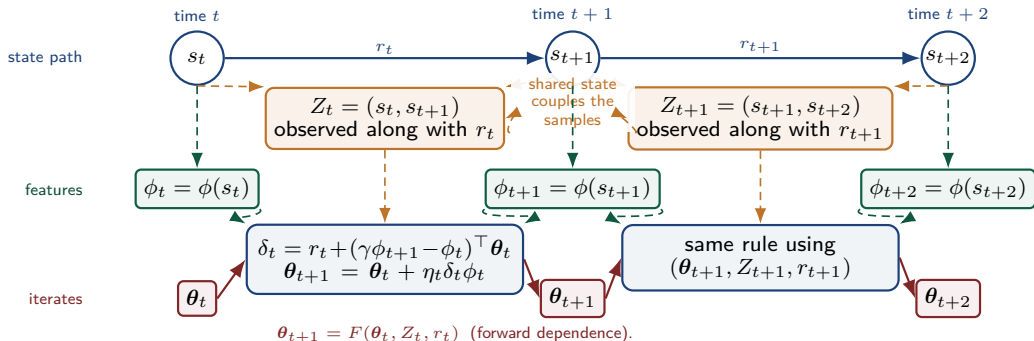
For a fixed policy μ , estimate the expected discounted return from state s with discount factor $\gamma \in (0, 1]$:

$$V^\mu(s) = \mathbb{E} \left[\sum_{k \geq 0} \gamma^k r_k \mid s_0 = s \right].$$

Setting II: Linear TD(0) with Markovian sampling

A fixed policy induces one **dependent** trajectory. Linear TD(0) combines the current iterate with the current transition sample:

$$\begin{aligned} Z_t &= (s_t, s_{t+1}), \\ V_{\theta}(s) &= \phi(s)^\top \theta, \quad \|\phi(s)\| \leq 1, \\ \delta_t(\theta_t) &= r_t + \gamma \phi(s_{t+1})^\top \theta_t - \phi(s_t)^\top \theta_t, \quad \theta_{t+1} = \theta_t + \eta_t \delta_t(\theta_t) \phi(s_t). \end{aligned}$$



Performance measure: TD potential and two regimes

For $v \in \mathbb{R}^{|\mathcal{S}|}$, let $\mathbf{D} := \text{diag}(\pi)$, where π is the stationary state distribution, and define

$$\|v\|_{\mathbf{D}}^2 := v^\top \mathbf{D} v = \sum_s \pi(s) v(s)^2.$$

We measure error using the TD potential

$$f(\boldsymbol{\theta}) = (1 - \gamma) \|\mathbf{V}_{\boldsymbol{\theta}} - \mathbf{V}_{\boldsymbol{\theta}^*}\|_{\mathbf{D}}^2.$$

Here, $\Phi = [\phi(s)^\top]_{s \in \mathcal{S}} \in \mathbb{R}^{|\mathcal{S}| \times d}$ is the feature matrix, and assume to have full column rank. Then f has curvature around the TD fixed point $\boldsymbol{\theta}^*$:

$$f(\boldsymbol{\theta}) - f(\boldsymbol{\theta}^*) \geq \omega \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|^2, \quad \omega := (1 - \gamma) \lambda_{\min}(\Phi^\top \mathbf{D} \Phi) > 0.$$

Two finite-time regimes

Robust rate	Fast rate
curvature-free	curvature-dependent
$\tilde{\mathcal{O}}(1/\sqrt{T})$	$\tilde{\mathcal{O}}(1/(\omega T))$

Why stability is hard under Markovian sampling

Two coupled difficulties.

- **Not i.i.d.:** $Z_t = (s_t, s_{t+1})$ is generated by the same trajectory that generated θ_t . Thus, in general,

$$\mathbb{E}[g(\theta_t, Z_t) \mid \mathcal{F}_{t-1}] \neq \bar{g}(\theta_t).$$

- **Self-amplification:**

$$\|g(\theta_t, Z_t)\| \leq r_\infty \phi_\infty + 2\phi_\infty^2 \|\theta_t\|.$$

A loose bound on $\|\theta_t\|$ makes the update bound loose, which feeds back into θ_{t+1} .

What prior work uses

- **Projection:** (Bhandari et al. 2018; Liu and Olshevsky 2021). Stable by design, but changes the TD algorithm and requires prior knowledge of a suitable projection radius.
- **Curvature / contractive stability:** (Srikant and Ying 2019; Patil et al. 2023; Samsonov et al. 2024; Mitra 2025; Chandak and Borkar 2025; Durmus et al. 2025; Li et al. 2026). Projection-free or lightly modified, but the rate, burn-in, or tuning can depend on the curvature.

Question: a single plain TD stepsize for both regimes?

Question

Can unprojected TD(0) achieve high-probability stability and simultaneous robust and fast rates without knowledge of ω ?

Best-of-both-worlds target

$$f(\bar{\theta}_T) - f(\theta^*) \leq \tilde{O}\left(\min\left\{\frac{1}{\sqrt{T}}, \frac{1}{\omega T}\right\}\right)$$

- If ω is small: still retain a $1/\sqrt{T}$ guarantee.
- If ω is large: automatically obtain the fast $1/(\omega T)$ rate.

Main results: stability and simultaneous rates

Use a single decaying stepsize

$$\eta_t = \frac{1}{c} \frac{1}{\tau_{\text{mix}}} \frac{1}{\sqrt{t+1} \log(t+3)}.$$

Let

$$R_{\text{base}} = \max \left\{ \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|, \|\boldsymbol{\theta}^*\|, \frac{r_\infty}{\phi_\infty} \right\}.$$

Define the weighted Polyak–Ruppert average

$$S_T = \sum_{t=1}^T \eta_{t-1}, \quad \bar{\boldsymbol{\theta}}_T = \frac{1}{S_T} \sum_{t=1}^T \eta_{t-1} \boldsymbol{\theta}_{t-1}.$$

Mixing assumption

Assume the transition chain $Z_t = (s_t, s_{t+1})$ is irreducible and aperiodic with stationary law π_Z . Then,

$$\max_{z \in \mathcal{Z}} \|P_Z^k(z, \cdot) - \pi_Z\|_{\text{TV}} \leq 4 \cdot 2^{-k/\tau_{\text{mix}}}, \quad k \geq 0.$$

Main theorem

For any sufficiently large c depending on δ , set $R_{\text{max}} = \rho(c, \delta) R_{\text{base}}$. With probability at least $1 - \delta$, for all $T \geq 1$,

$$\sup_{t \geq 0} \|\boldsymbol{\theta}_t\| \leq R_{\text{max}}, \quad f(\bar{\boldsymbol{\theta}}_T) - f(\boldsymbol{\theta}^*) \leq \tilde{\mathcal{O}} \left(R_{\text{max}}^2 \min \left\{ \frac{\tau_{\text{mix}}^2}{\omega T}, \frac{\tau_{\text{mix}}}{\sqrt{T}} \right\} \right).$$

Stability proof I: make the loop self-bounding

Use induction on the past event

$$\text{IH}_{t-1} := \left\{ \max_{0 \leq i < t} \|\boldsymbol{\theta}_i\| \leq R_{\max} \right\}.$$

Since $\|\boldsymbol{\theta}_t\| \leq \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\| + \|\boldsymbol{\theta}^*\|$, we can use $\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|$ to control $\|\boldsymbol{\theta}_t\|$. Expanding $\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2$ gives

$$\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 = \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|^2 + \sum_{i=0}^{t-1} \eta_i^2 \|\mathbf{g}_i\|^2 + 2 \sum_{i=0}^{t-1} \eta_i \langle \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle + 2 \sum_{i=0}^{t-1} \eta_i h_i(Z_i),$$

where

$$\mathbf{g}_i := \mathbf{g}(\boldsymbol{\theta}_i, Z_i), \quad h_i(z) := \langle \bar{\mathbf{g}}(\boldsymbol{\theta}_i, z) - \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle.$$

The mean drift is helpful:

$$\langle \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle = -[f(\boldsymbol{\theta}_i) - f(\boldsymbol{\theta}^*)] \leq 0.$$

Under IH_{t-1} , the quadratic term is controlled because $\sum_i \eta_i^2 < \infty$. The only hard term is the Markov bias

$$B_t := \sum_{i=0}^{t-1} \eta_i h_i(Z_i).$$

Stability proof II: use the Poisson equation for the Markov bias

Freeze θ_i . Since $\mathbb{E}_{Z \sim \pi_Z}[h_i(Z)] = 0$, solve the Poisson equation

$$u_i - P_Z u_i = h_i, \quad u_i(z) = \sum_{\ell=0}^{\infty} (P_Z^\ell h_i)(z), \quad \|u_i\|_\infty \leq 16\tau_{\text{mix}} \|h_i\|_\infty.$$

Then each Markov-noise term decomposes into

$$h_i(Z_i) = \underbrace{u_i(Z_{i+1}) - (P_Z u_i)(Z_i)}_{\text{martingale difference}} + \underbrace{u_i(Z_i) - u_i(Z_{i+1})}_{\text{telescoping remainder}}.$$

Under $\mathcal{IH}_{t-1}$,

$$\|h_i\|_\infty \lesssim R_{\text{max}}^2, \quad \|u_i\|_\infty \lesssim \tau_{\text{mix}} R_{\text{max}}^2.$$

Two key implications

- The first term is controlled by Pinelis' inequality for bounded differences.
- The second term is rewritten using Abel summation; the time-varying functions u_i are controlled by Lipschitz continuity and $\theta_i - \theta_{i-1} = \eta_{i-1} g_{i-1}$.

Stability proof III: close the bootstrap

The previous bounds give

$$\|\boldsymbol{\theta}_t\|^2 \leq \left(4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2}\right) R_{\text{base}}^2.$$

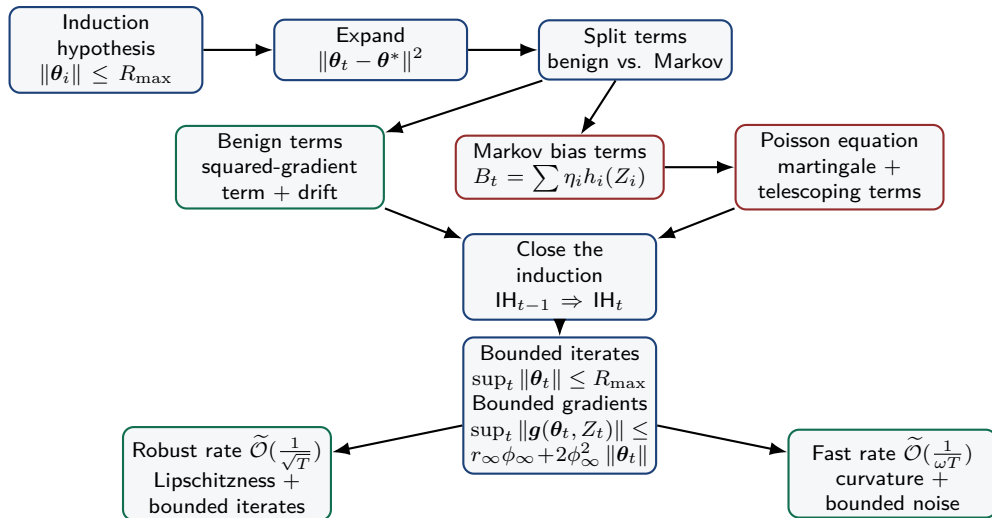
To close the induction, it is enough that

$$4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2} \leq \rho^2.$$

Many pairs (c, ρ) close the induction

- As $c \rightarrow \infty$, the admissible radius factor satisfies $\rho \downarrow 2$.
- There is a minimum requirement $c > c_{\min}(\delta)$.
- Larger c : smaller stepsize, smaller radius bound.
- Smaller c above the threshold: larger stepsize, larger radius bound.

Proof roadmap: stability first, rates second



Conclusion

Main message

A single ω -agnostic stepsize suffices for plain linear TD(0) under Markovian sampling to achieve a $\tilde{\mathcal{O}}\left(\min\left\{\frac{1}{\sqrt{T}}, \frac{1}{\omega T}\right\}\right)$ rate without projection.

- **Algorithmic takeaway:** no projection radius and no tuning based on the curvature/eigenvalue parameter ω .
- **Proof takeaway:** The TD gradient structure makes self-bounding induction possible and gives implicit boundedness.

Open problem: Can we still get a reasonable best-of-both-worlds rate with a τ_{mix} -agnostic stepsize?

- The current stepsize is $\eta_t = \frac{1}{c} \frac{1}{\tau_{\text{mix}}} \frac{1}{\sqrt{t+1} \log(t+3)}$, but τ_{mix} is usually unknown in practice.
- A mixing-free stepsize is available: $\eta_t = \frac{1}{c} \frac{1}{\sqrt{t+1} \log^2(t+3)}$, but its constants have doubly exponential dependence on τ_{mix} .

Thank you! Any questions?

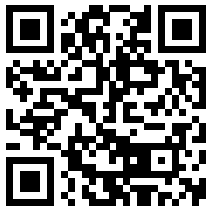
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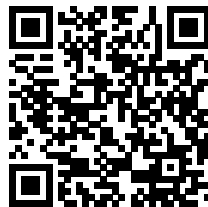
We also have a poster!



Slides



Paper



Personal website

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